MATH 2431 | Honors Probability | Spring 2026
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Problem Set 3

Submission:
Friday, 06/03/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.
Only problems marked with $(*)$ will be graded.

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1. **$(*)$ The Monty Hall problem** [4 Points]

   In a game show, you have to choose one of three doors. Behind one door is a new car, and behind the other two doors are goats. After you make your choice, the host reveals one of the other doors, which always has a goat behind it. You are then given the option to switch your choice to the remaining unopened door.

   (a) Given that the host has opened a door revealing a goat, should you switch your choice to increase the chances of winning the car? Please explain your reasoning.

   (b) Let's call the two goats Bill and Nan. Given that the host has opened a door revealing a goat, and shows Bill with a probability of b, following the strategy in (a), what is the probability of winning the car, $i.e.$ winning a car given Bill is revealed? [Note: This means that if the host can reveal either Bill or Nan (a door with the car behind it is initially chosen), the host will reveal Bill with probability b.]

   (c) If the host opens a door randomly, which strategy would you take to maximize your chances of winning the car? [Hint: Consider two cases separately, one of which is trivial.]

2. **On the definition of countable sums**

   Recall that for a countable set Ω and a function $f : \Omega \to [0, \infty)$, we have defined the meaning of

   $$\sum_{\omega \in \Omega} f(\omega) = \sup \left\{ \sum_{\omega \in F} f(\omega) \ : \ F \subseteq \Omega, \ F \text{ finite} \right\}$$

   (with $\sum_{\omega \in \varnothing} f(\omega) = 0$) in Definition 3.1. It is clear that this definition is consistent with the definition of finite sums when Ω is finite. Show that the other two claims in Remark 3.2 hold:

   (a) Suppose in addition that Ω is infinite. Then for an enumeration of $\Omega = \{\omega_1, \omega_2, \dots\}$ (see Remark 3.2(ii)),

   $$\sum_{\omega \in \Omega} f(\omega) = \sum_{j=1}^{\infty} f(\omega_j) = \lim_{N \to \infty} \sum_{j=1}^{N} f(\omega_j).$$

   Conclude that the limit on the right-hand side does not depend on the choice of the enumeration.

   (b) If $(A_j)_{j \in \mathbb{N}}$ are pairwise disjoint with $\Omega = \bigcup_{j \in \mathbb{N}} A_j$, then

   $$\sum_{\omega \in \Omega} f(\omega) = \sum_{j=1}^{\infty} \sum_{\omega \in A_j} f(\omega).$$

3. **$(*)$ Discrete distributions** [4 Points]

   For (a) and (b), first determine which probability distribution you are using.

(a) We roll a fair die until the first time we observe the number 6. What is the probability that we need more than 9 attempts?

(b) In the production of a certain kind of computer chip defects occur independently with probability 0.01. How likely is it that there are at least 2 defects in a batch of 100 computer chips? Give the exact probability and compare it to the approximation with a Poisson distribution.

Show that the following approximation of probability mass functions holds:

(c) The probability mass function of a hypergeometric distribution $H(N, M, n)$ with parameters $N, M, n \in \mathbb{N}$ converges to the probability mass function of a binomial distribution $Bin(n, p)$ as $N, M \to \infty$ if $\frac{M}{N} \to p$.

Consider the following thinning of a Poisson distribution:

(d) Assume that the number of penalty kicks during a football match is Poisson distributed with some parameter $\lambda > 0$. Also assume that every penalty kick results in a goal with the same probability $p \in (0, 1)$, independently of all others. Determine the distribution of the number of goals obtained via a penalty kick.

Hint: For $k \in \mathbb{N}_0$, calculate the probability that there are exactly k goals obtained via a penalty kick.

4. **n-step simple random walk and the ballot theorem**

Let $n \in \mathbb{N}$. We define the set

$$\Omega_n = \{(\omega_j)_{j=0}^n \in \mathbb{Z}^n \,:\, \omega_0 = 0, |\omega_j - \omega_{j-1}| = 1, \text{for all } j = 1, ..., n\}.$$

We call the discrete uniform distribution on $(\Omega_n, \mathscr{P}(\Omega_n))$ the (law of the) n-step simple random walk starting from 0, and abbreviate it by P_0.

(a) Calculate $|\Omega_n|$. Explain in words why an outcome $\omega \in \Omega_n$ can be viewed as a trajectory of a simple random walk.

(b) Consider for $k \in \mathbb{Z}$ the event

$$A_k = \{\omega \in \Omega_n \,:\, \omega_n = k\}.$$

Calculate $P_0[A_k]$. What is the interpretation of the event A_k?

We consider now the following application: Suppose when voting for a certain motion A, $n \in \mathbb{N}$ ballots are cast. After all ballots are cast, they are counted one by one, and the result "for A" and "against A" is noted. We assume that it is known that A passes with $k \in \{1, ..., n\}$ more "for A" votes than "against A" votes. We are interested in the probability that during the entire counting process, there are always (except at the beginning, when no ballots have been counted) more "for A" votes than "against A" votes.

(c) Argue that this corresponds to calculating

$$P_0[\{\omega \in \Omega_n \,:\, \omega_j > 0, 1 \leq j \leq n\} | A_k],$$

and calculate the previous expression.
Hint: You need to find the number of $(n-1)$-step paths from 1 to k that stay strictly positive. To do this, subtract from the total number of *all* $(n-1)$-step paths from 1 to k the number of paths that attain a value ≤ 0.